

Zhoujun (Jorge) Cheng

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Education

University of California San Diego , CA, USA	Sept 2024 – Present
• Ph.D. in Computer Science and Engineering	
• Advisor: Prof. Zhiting Hu	
Shanghai Jiao Tong University , Shanghai, China	Sept 2021 – Mar 2024
• M.E. in Computer Science	
• Advisor: Prof. Fan Cheng	
Shanghai Jiao Tong University , Shanghai, China	Sept 2017 – Jun 2021
• B.E. in Computer Science (IEEE honor class)	
• Advisor: Prof. Cewu Lu	

Experience

MBZUAI, Institute of Foundation Models – Sunnyvale, CA	Mar 2025 – Present
• Research Intern, <i>Advisor: Dr. Hector Liu</i>	
• Topics: Post-training (supervised fine-tuning and reinforcement learning) for the K2-Think-32B/70B model.	
MBZUAI, Institute of Foundation Models – Sunnyvale, CA (Virtual)	Oct 2024 – Mar 2025
• Research Assistant, <i>Advisor: Dr. Hector Liu</i>	
• Topics: Building the scaling law for a large-scale (70B) LLM pretrained from scratch.	
The University of Hong Kong, XLang Group – Hong Kong, China	Apr 2022 – Jan 2024
• Research Assistant, <i>Advisor: Prof. Tao Yu</i>	
• Topics: Binding tool calls to LLMs; digital agents in computer use and coding;	
Microsoft Research Asia, DKI Group – Beijing, China	Dec 2020 – Apr 2022
• Research Intern, <i>Advisor: Haoyu Dong</i>	
• Topics: Hierarchical table and spreadsheet understanding.	
Shanghai Jiao Tong University, MVIG Group – Shanghai, China	Jul 2019 – Jan 2020
• Research Intern, <i>Advisor: Prof. Cewu Lu</i>	
• Topics: Dense semantic matching of 3D objects.	

Preprints and Publications

Works that I am a primary author on are **highlighted**; * indicates equal contribution.

[1] Revisiting Reinforcement Learning for LLM Reasoning from A Cross-Domain Perspective

Zhoujun Cheng*, Shibo Hao*, Tianyang Liu*, Fan Zhou, Yutao Xie, Feng Yao, Yuexin Bian, Yonghao Zhuang, Nilabjo Dey, Yuheng Zha, Yi Gu, Kun Zhou, Yuqi Wang, Yuan Li, Richard Fan, Jianshu She, Chengqian Gao, Abulhair Saparov, Haonan Li, Taylor W. Killian, Mikhail Yurochkin, Zhengzhong Liu, Eric P. Xing, Zhiting Hu

Preprint, in submission to NeurIPS 2025 Datasets & Benchmarks

[2] MegaMath: Pushing the Limits of Open Math Corpora

Fan Zhou*, Zengzhi Wang*, Nikhil Ranjan, **Zhoujun Cheng**, Liping Tang, Guowei He, Zhengzhong Liu, Eric P. Xing

COLM 2025

[3] OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments

Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh Jing Hua, **Zhoujun Cheng**, Dongchan Shin, Fangyu Lei, Yitao Liu, Yiheng Xu, Shuyan Zhou, Silvio Savarese, Caiming Xiong, Victor Zhong, Tao Yu

NeurIPs 2024 Datasets & Benchmarks (benchmarked by OpenAI & Anthropic)

[4] OpenAgents: An Open Platform for Language Agents in the Wild

Tianbao Xie*, Fan Zhou*, **Zhoujun Cheng***, Peng Shi*, Luoxuan Weng*, Yitao Liu*, Toh Jing Hua, Junning Zhao, Qian Liu, Che Liu, Leo Z. Liu,

Yiheng Xu, Hongjin Su, Dongchan Shin, Caiming Xiong, Tao Yu

COLM 2024 ([Github 4K stars](#), [7k demo users](#))

[5] What Are Tools Anyway? A Survey from the Language Model Perspective

Zhiruo Wang, Zhoujun Cheng, Hao Zhu, Daniel Fried, Graham Neubig

COLM 2024

[6] Lemur: Harmonizing Natural Language and Code for Language Agents

Yiheng Xu*, Hongjin Su*, Chen Xing*, Boyu Mi, Qian Liu, Weijia Shi, Binyuan Hui, Fan Zhou, Yitao Liu, Tianbao Xie, Zhoujun Cheng, Siheng Zhao, Lingpeng Kong, Bailin Wang, Caiming Xiong, Tao Yu

ICLR 2024 ([Spotlight](#))

[7] Binding Language Models in Symbolic Languages

Zhoujun Cheng*, Tianbao Xie*, Peng Shi, Chengzu Li, Rahul Nadkarni, Yushi Hu, Caiming Xiong, Dragomir Radev, Mari Ostendorf, Luke Zettlemoyer, Noah A. Smith, Tao Yu

ICLR 2023 ([Spotlight](#))

[8] Batch Prompting: Efficient Inference with Large Language Model APIs

Zhoujun Cheng, Jungo Kasai, Tao Yu

EMNLP 2023 Industry Track

[9] TaCube: Pre-computing Data Cubes for Answering Numerical-Reasoning Questions over Tabular Data

Fan Zhou, Mengkang Hu, Haoyu Dong, Zhoujun Cheng, Shi Han, Dongmei Zhang

EMNLP 2022 ([Oral](#))

[10] HiTab: A Hierarchical Table Dataset for Question Answering and Natural Language Generation

Zhoujun Cheng*, Haoyu Dong*, Zhiruo Wang*, Ran Jia, Jiaqi Guo, Yan Gao, Shi Han, Jian-Guang Lou, Dongmei Zhang

ACL 2022

[11] FORTAP: Using Formulae for Numerical-Reasoning-Aware Table Pretraining

Zhoujun Cheng*, Haoyu Dong*, Ran Jia, Pengfei Wu, Shi Han, Fan Cheng, Dongmei Zhang

ACL 2022

[12] Table Pre-training: A Survey on Model Architectures, Pretraining Objectives, and Downstream Tasks

Haoyu Dong, Zhoujun Cheng, Xinyi He, Mengyu Zhou, Anda Zhou, Fan Zhou, Ao Liu, Shi Han, Dongmei Zhang

IJCAI 2022 Survey Track

[13] KeypointNet: A Large-scale 3D Keypoint Dataset Aggregated from Numerous Human Annotations

Yang You, Yujing Lou*, Chengkun Li*, Zhoujun Cheng, Liangwei Li, Lizhuang Ma, Cewu Lu, Weiming Wang

CVPR 2020

[14] Human Correspondence Consensus for 3D Object Semantic Understanding

Yujing Lou*, Yang You*, Chengkun Li*, Zhoujun Cheng, Liangwei Li, Lizhuang Ma, Weiming Wang, Cewu Lu

ECCV 2020

Awards & Scholarships

Shanghai Outstanding Graduates	2021
MSRA Stars of Tomorrow	2021
2nd Place, IBM “Call for Code” Hackathon, Shanghai	2019
Shanghai Jiao Tong University, Zhiyuan Honor Scholarship (top 5%)	2018 – 2020
National Scholarship	2018

Services & Teaching

Reviewer: NeurIPs 2025, ICLR 2025, ACL Rolling Review(ARR) 2022-2024, NAACL 2024, ACL 2024, EACL 2024, COLING 2024, CCL 2022-2023

Teaching Assistant:

- Introduction to Programming, Shanghai Jiao Tong University
- Introduction to Machine Learning, Shanghai Pinghe High School

Open-Source

OpenAgents (Primary Contributor) 4.4K stars

- An agent platform for using, hosting and building language agents, covering:
 - Data Agent: data analysis by generating Python/SQL code, augmented with data tools
 - Plugins Agent: integrating 200+ third-party plugins to handle daily life scenarios
 - Web Agent: exploring websites automatically as a browser extension

veRL: A reinforcement learning framework for LLM 12.4K stars

- Fixed a loss mask bug in veRL SFT code that caused the model's first generated token to be random.

LLM-Reasoners: A library of advanced reasoning algorithms (e.g., MCTS) for LLM 2.2K stars

- Implemented the tree search algorithm in a web agent scenario to improve performance.

TxT360: A large-scale open pretraining corpus with 5T globally-deduped tokens 1.76M downloads

- Parsed and cleaned the academic paper data.

Skills

Programming: Python, C/C++, SQL, Java, Javascript, HTML, ~~HTML~~ $\text{\LaTeX}$

LLM Training/Inference: PyTorch, veRL, Megatron-LM, (PyTorch) FSDP, vLLM

General Tools: Docker, Git, Slurm, MongoDB, Redis, Flask

Languages: English: Fluent, Mandarin: Native